

SAP CPI Project Documentation

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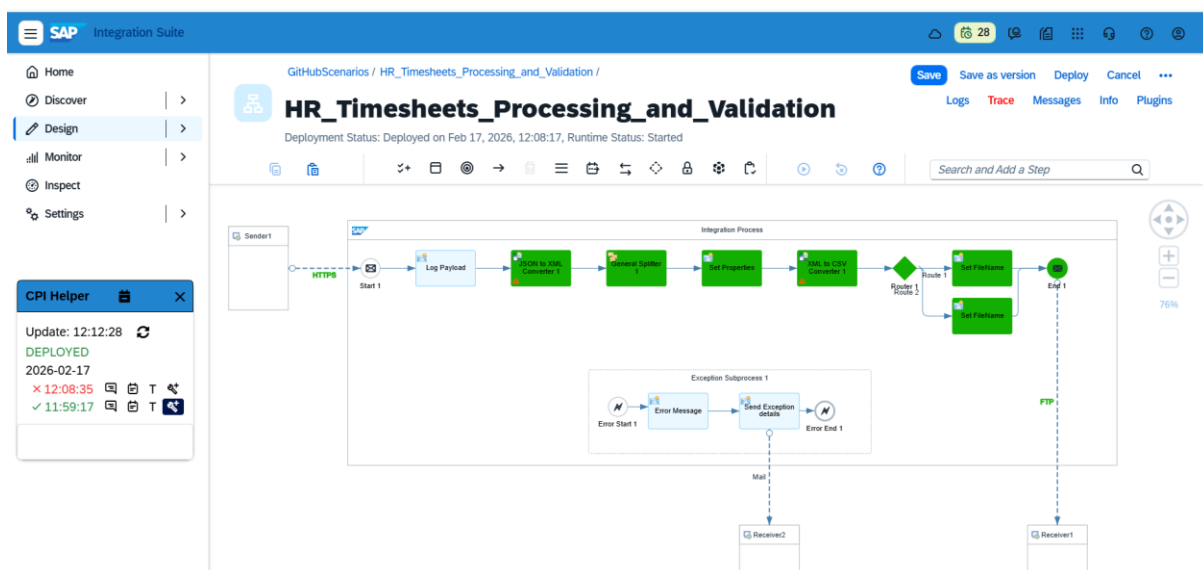
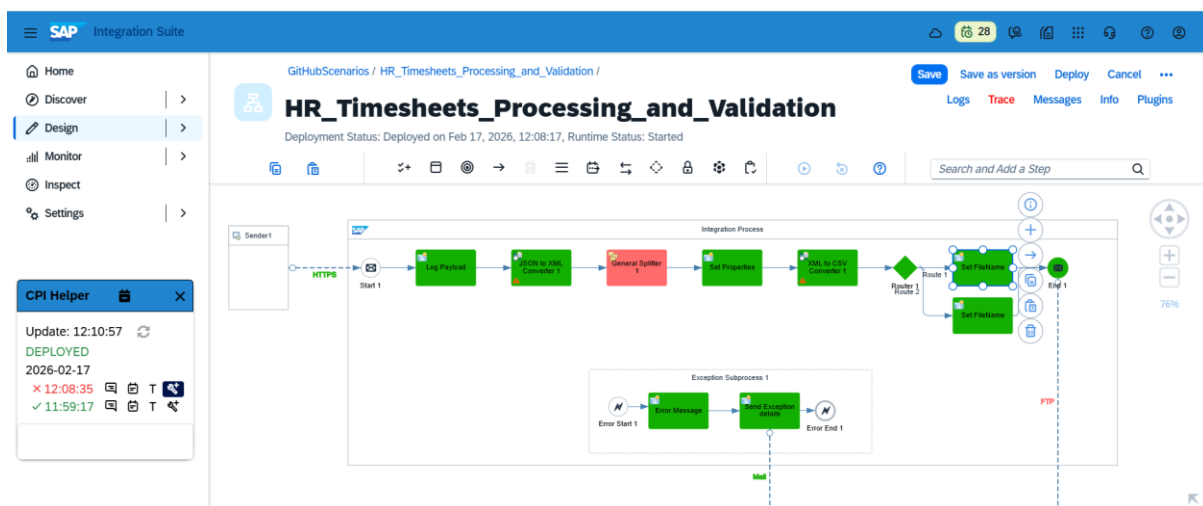
Date:10/12/2026

Scenario: HR_Timesheets_Processing_and_Validation

1.Scenario Overview

This integration handles employee timesheet submissions from a web application. Timesheet records are validated, converted, and stored in the HR system in CSV format using region-specific folders. It demonstrates the use of HTTP Adapter, Splitter, Filter, Router, Format Conversion, and File Adapter, suitable for automating and validating HR operations.

2.Architechture Diagram



3. Integration Flow

Steps

1. **HTTP Sender Adapter** receives batch JSON data

The screenshot shows the configuration for an HTTP Sender Adapter. The 'Connection' tab is selected. Under 'REQUEST PROCESSING', the 'Address' is set to '/employee/timesheets', 'Authorization' is 'User Role', 'User Role' is 'ESBMessaging.send', and 'CSRF Protected' is checked.

HTTPS		
General	Connection	Conditions
REQUEST PROCESSING		
Address:	/employee/timesheets	
Authorization:	User Role	
User Role:	ESBMessaging.send	
CSRF Protected:	☒	

2. **JSON to XML Converter** prepares structure for processing

3. **XML Splitter** breaks down entries into individual records

The screenshot shows the configuration for an XML Splitter. The 'Processing' tab is selected. 'Expression Type' is 'XPath', 'XPath Expression' is '/root/Timesheets', 'Grouping' is empty, 'Streaming' is checked, 'Parallel Processing' is unchecked, and 'Stop on Exception' is unchecked.

General Splitter	
General	Processing
Expression Type:	XPath
XPath Expression:	/root/Timesheets
Grouping:	
Streaming:	☒
Parallel Processing:	☐
Stop on Exception:	☐

4. **Content Modifier** adds metadata (e.g., company name, processed timestamp) by **Setting Properties**

The screenshot shows the configuration for a Content Modifier. The 'Exchange Property' tab is selected. A table lists properties to be added to the message.

Content Modifier					
Properties:					
Action	Name	Source Type	Source Value	Data Type	Default Value
Create	EmployeeID	XPath	//EmployeeID	java.lang.String	
Create	Date	XPath	replace(//Date, ',', ' ')	java.lang.String	
Create	Hours	XPath	//Hours	java.lang.String	
Create	CompanyName	Constant	Infosys		
Create	ProcessedDate	Expression	\$(date:now)		

5. Router separates valid and invalid records using exchange property

The screenshot shows the 'Router' configuration window with the 'Processing' tab selected. It displays error handling and routing condition settings.

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Route 1	<code>\${property.Hours} <= '12'</code>	☐
2	Route 2		☒

6. XML to CSV Converter formats data

7. File Receiver Adapter writes valid CSVs to /hr/valid/, invalid ones to /hr/invalid/

The screenshot shows the 'FTP' configuration window with the 'Target' tab selected. It displays file access and connection parameters.

FILE ACCESS PARAMETERS

Directory: `/hr/${property.directory}`

File Name: `${property.EmployeeID}_${property.Date}.csv`

Append Timestamp: ☐

CONNECTION PARAMETERS

Address: `ftp.drivehq.com`

Proxy Type: Internet

Encryption: Explicit FTPS

Credential Name: `FTP_Connect`

Timeout (in ms): `10000`

Maximum Reconnect Attempts: `3`

8. Sample Input JSON (from Web App)

```
{
  "Timesheets": [
    { "EmployeeID": "E123", "Date": "2025-07-16", "Hours": 9 },
    { "EmployeeID": "E456", "Date": "2025-07-16", "Hours": 14 }
  ]
}
```

9. Converted XML (after JSON to XML)

<Timesheets>

<Entry>

<EmployeeID>E123</EmployeeID>

<Date>2025-07-16</Date>

<Hours>9</Hours>

</Entry>

<Entry>

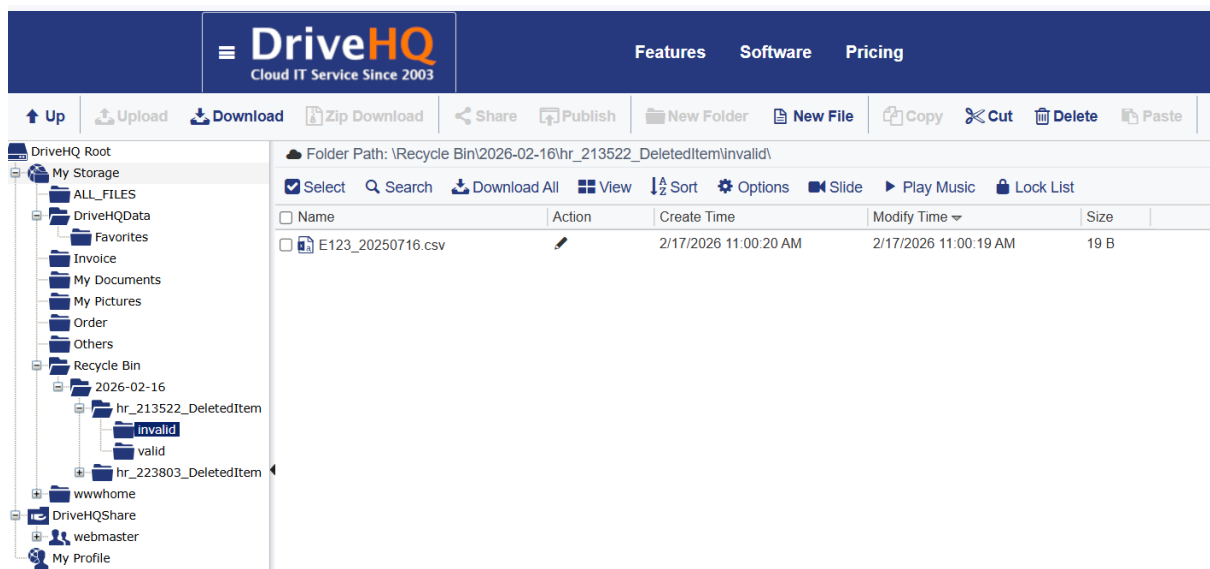
<EmployeeID>E456</EmployeeID>

<Date>2025-07-16</Date>

<Hours>14</Hours>

</Entry>

</Timesheets>

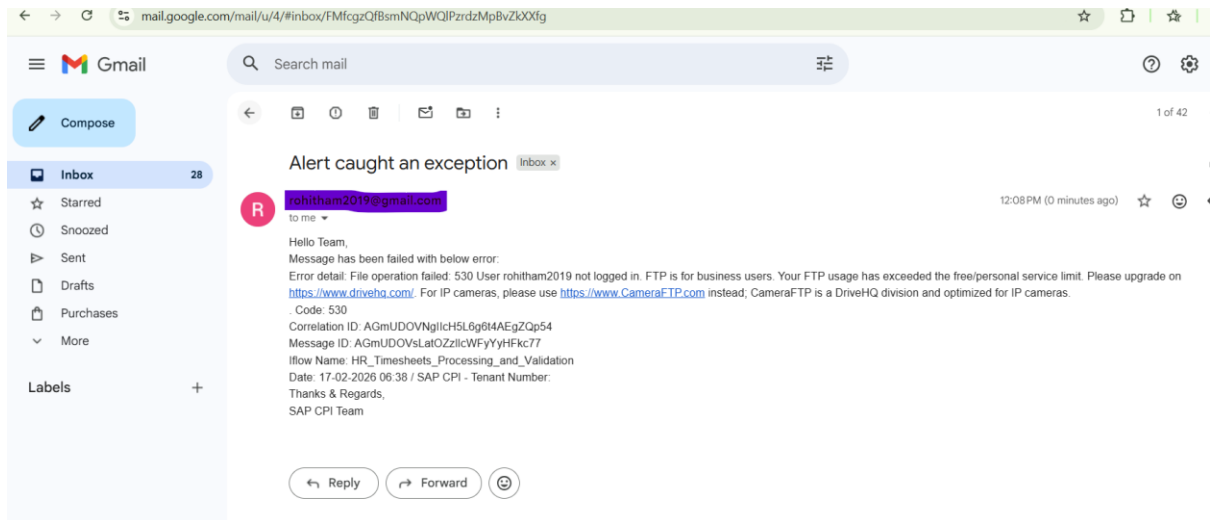


Files stored in HR system

10. Exception Handling

- Any transformation or file-write error is:

- Alert sent via **Gmail Adapter** with employee ID and error details



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Iflow Package



HR_Timesheets_Processing_and_Validation.zip